

CIPD 360 ERP: A Scalable Academic Management System with Automated Attendance and Feedback Intelligence

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Student Declaration

We hereby declare that the work presented in the report entitled “**CIPD 360 ERP: A Scalable Academic Management System with Automated Attendance and Feedback Intelligence**” submitted by us for the partial fulfilment of the requirements for the degree of *B.Tech. in Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of our work carried out under the guidance of **Prof. Anuj Grover**. Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

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Abstract

This report presents the design and development of CIPD 360 ERP, a scalable academic management system developed for the Centre for Intelligent Product Development (CiPD) program at IIIT Delhi. The system aims to address inefficiencies in traditional academic workflows by providing an integrated platform for scheduling, attendance tracking, feedback collection, and reporting.

A key innovation of the system is the implementation of a WiFi based automated attendance mechanism, which leverages network level device detection to infer student presence without requiring manual input or additional hardware. This approach was selected after evaluating multiple alternatives, including Bluetooth based detection, RFID systems, and face recognition techniques, with a focus on scalability and practicality.

The system further incorporates a session driven feedback mechanism, enabling structured and timely collection of student feedback, along with analytics to derive actionable insights. A role based architecture ensures that administrators, faculty, and students have access to personalized dashboards and relevant functionalities.

In addition, the platform integrates scheduling, attendance, and feedback into a unified workflow, allowing automation of key processes such as attendance computation, feedback rollout, and notification generation. This reduces administrative overhead and enhances operational efficiency.

The proposed system demonstrates the feasibility of building a centralized, data driven academic ERP tailored for dynamic educational environments, with potential for scalability and adoption across similar programs.

Keywords: Academic ERP, Attendance Automation, WiFi Detection, Feedback Analytics, Scheduling System, Educational Technology

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Chapter 1

Introduction

1.1 Background

In modern academic institutions, efficient management of administrative and academic processes is critical for ensuring smooth operation and high quality learning outcomes. However, many institutions still rely on partially manual or loosely integrated systems for managing core activities such as attendance tracking, scheduling, feedback collection, and academic reporting.

Traditional attendance systems often involve manual roll calls or basic digital entry, both of which are prone to errors, proxy attendance, and inefficiencies. Similarly, scheduling is frequently handled through static timetables with limited flexibility, leading to conflicts and lack of real time updates. Feedback collection mechanisms, where implemented, are often inconsistent and fail to provide actionable insights due to low participation or poor structure.

Furthermore, the absence of centralized systems results in fragmented data across different processes. This makes it difficult for administrators and faculty to obtain a comprehensive view of student engagement, faculty workload, and overall academic performance. As institutions scale, these inefficiencies become more pronounced, highlighting the need for an integrated and automated solution.

The CiPD (Centre for Intelligent Product Development) program at IIIT Delhi operates in a dynamic and fast paced academic environment, where multiple sessions, instructors, and student interactions occur daily. This setting demands a system that not only manages operations efficiently but also provides real time insights and automation to support decision making.

1.2 Problem Statement

The primary problem addressed in this project is the lack of a unified, scalable, and automated system to manage academic operations within the CiPD program.

Specifically, the following challenges were identified:

- **Inefficient Attendance Tracking:** Existing attendance methods are either manual or semi digital, making them susceptible to inaccuracies, proxy attendance, and delayed processing.
- **Lack of Real Time Scheduling Management:** Scheduling of sessions is not dynamically managed, leading to potential conflicts, lack of visibility, and difficulty in handling changes.
- **Inadequate Feedback Collection and Analysis:** Feedback systems, if present, are not tightly integrated with sessions, resulting in low response rates and limited actionable insights.
- **Fragmented Data and Limited Analytics:** Data related to attendance, feedback, and academic activities is not consolidated, restricting the ability to perform meaningful analysis and generate reports.
- **Absence of Automation in Academic Workflows:** Key workflows such as attendance processing, feedback rollout, and notifications are not automated, increasing administrative overhead.

These challenges collectively reduce operational efficiency, limit data driven decision making, and affect the overall academic experience for students, faculty, and administrators.

The objective is to design a scalable prototype that demonstrates the feasibility and effectiveness of an integrated academic ERP system, with potential for further expansion and enhancement in future iterations.

Chapter 2

Research, Ideation and Continuous Evaluation

2.1 Stakeholder Research and Problem Discovery

The project began with a user centric research phase aimed at understanding the real world challenges within the CiPD program. Interactions were conducted with students, faculty members, and administrative staff to identify pain points and validate potential solutions.

Students highlighted issues with manual attendance systems, describing them as inconvenient and prone to proxy marking and lack of transparency. They also emphasized the absence of a centralized platform for accessing schedules, resources, and tracking participation.

Faculty members pointed out difficulties in managing schedules, monitoring engagement, and collecting structured feedback. Administrative stakeholders faced challenges in consolidating data for reporting and decision making.

Based on these discussions, the key problem areas identified were:

- Inefficient and unreliable attendance tracking
- Lack of a centralized integrated system
- Inconsistent feedback mechanisms
- Limited access to real time analytics

An iterative validation process was followed, where early ideas were continuously discussed with stakeholders to ensure alignment with actual user needs.

2.2 Exploration of Attendance Solutions

Attendance tracking emerged as the most critical component, leading to the evaluation of multiple approaches:

- Bluetooth based systems relied on proximity detection but were unreliable due to device settings and inconsistent discoverability.
- RFID systems offered accuracy but required additional hardware, making them costly and less scalable.
- Face recognition approaches using computer vision were explored but faced challenges related to accuracy, computational overhead, and privacy concerns.

After evaluating these options, Wi Fi based detection was selected as the most practical solution. It leverages existing infrastructure to detect connected devices and infer presence.

The choice was driven by:

- No additional hardware requirement
- Scalability for large classrooms
- Passive and non intrusive operation
- Compatibility with existing systems

This evaluation ensured that the final approach was both feasible and scalable.

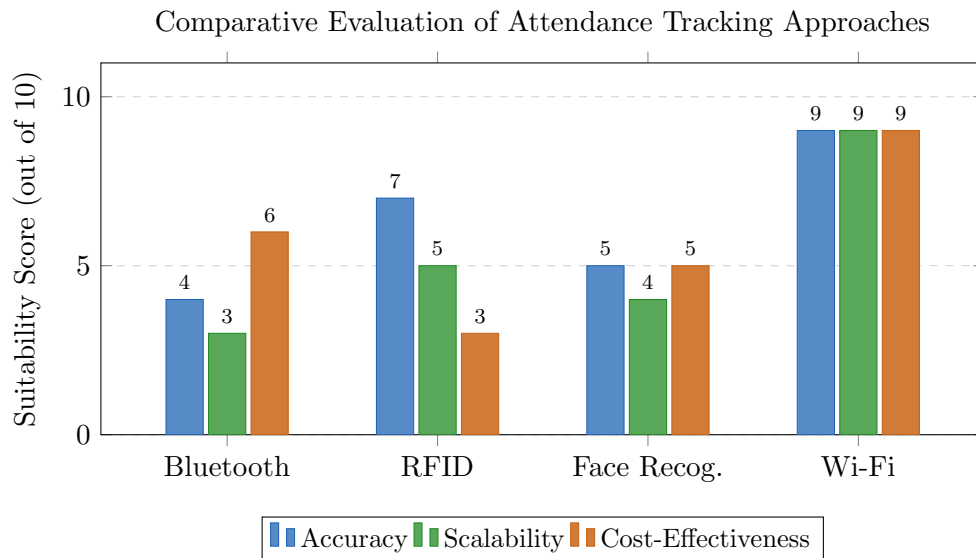


Figure 2.1: Comparative evaluation of four candidate attendance tracking approaches

2.3 Network Level Experimentation and Constraints

Following the selection of Wi Fi based attendance, experiments were conducted to test feasibility within the institute’s network. Initial trials using the existing network demonstrated that connected devices could be detected using network scanning techniques.

However, restricted access and permission limitations prevented continuous monitoring on the institutional network. To address this, a dedicated router was deployed within the CiPD classroom, creating a controlled environment for reliable device detection.

Attempts were also made to enhance the router’s capabilities using custom firmware (OpenWRT), but hardware limitations restricted further customization.

These experiments highlighted key practical constraints such as network permissions, infrastructure limitations, and hardware compatibility, which influenced the final system design.

2.4 Continuous Feedback and Iterative Development

The system was developed using an iterative approach, with regular demonstrations to stakeholders for feedback. This allowed continuous refinement of features such as dashboards, attendance tracking, and feedback interfaces.

Key improvements based on feedback included:

- Enhanced notification mechanisms for timely updates
- Improved visualization of attendance data
- Focus on student wise analytics instead of only aggregated views

This feedback driven approach ensured that the system remained aligned with user expectations and practical requirements while minimizing major redesign efforts.

Chapter 3

System Overview

3.1 Overview of CIPD 360 ERP

CIPD 360 ERP is a web based system that provides a centralized interface for managing academic activities. It replaces fragmented and manual processes with a cohesive platform that integrates multiple functionalities into a single system.

The system is built with the following goals:

- To provide a unified platform for all stakeholders
- To reduce manual intervention in routine academic processes
- To enable real time access to data and insights
- To support scalability for future expansion

By consolidating scheduling, attendance, feedback, and reporting into one system, the platform ensures consistency and reduces redundancy across operations.

3.2 Key Features Overview

The system incorporates several key features that collectively support academic management:

- **Scheduling System:** Enables administrators to create and manage academic sessions, with support for conflict detection and dynamic updates.
- **Automated Attendance System:** Uses WiFi based detection to track student presence, reducing dependency on manual attendance methods.
- **Feedback System:** Provides session based feedback collection with structured forms and analytical insights.

- **Dashboard and Reporting:** Offers role based dashboards with visual analytics for monitoring attendance, feedback, and session activity.
- **Content and Assignment Management:** Supports sharing of learning materials and management of assignments within the platform.

These features are tightly integrated, allowing smooth data flow and enabling automation across workflows.

3.3 User Roles and Responsibilities

The system is designed around 2 primary user roles, each with distinct responsibilities and access levels:

- **Administrator:** Responsible for managing the overall system, including scheduling sessions, monitoring attendance, generating reports, and configuring system settings. As well as managing faculties and other teaching staff.
- **Student:** Responsible for attending sessions, accessing course materials, submitting feedback, and tracking personal attendance and academic progress.

3.4 System Workflow

The CIPD 360 ERP follows a structured workflow that connects different modules into a cohesive pipeline:

1. **Session Scheduling:** Administrators create and manage sessions, defining details such as timing, faculty, and venue.
2. **Attendance Tracking:** During the session, the system monitors connected devices on the CiPD network and records presence data.
3. **Attendance Processing:** After the session, detection data is processed to compute attendance records and participation scores.
4. **Feedback Rollout:** Once a session is completed, feedback forms are automatically made available to students.
5. **Feedback Collection and Analysis:** Students submit feedback, which is aggregated and analysed for insights.
6. **Reporting and Visualization:** Data from all modules is consolidated and presented through dashboards and reports.

This workflow ensures that academic activities are not isolated but part of a continuous, automated process.

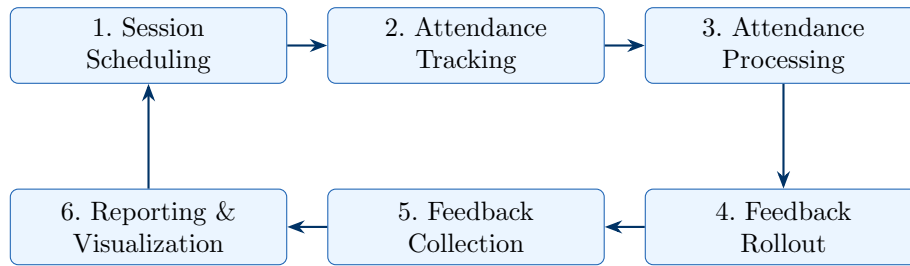


Figure 3.1: End-to-end operational workflow of CIPD 360 ERP

3.5 System Characteristics

The system is designed with several important characteristics:

- **Modularity:** Each module (attendance, scheduling, feedback, etc.) operates independently but integrates with others.
- **Scalability:** The architecture allows the system to handle increasing numbers of users and sessions.
- **Automation:** Key workflows such as attendance computation and feedback rollout are automated.
- **Real Time Data Access:** Users have access to up to date information through dynamic dashboards.
- **User Centric Design:** Interfaces are tailored to the needs of different roles.

These characteristics ensure that the system remains robust, flexible, and efficient.

Chapter 4

System Architecture and Implementation

4.1 High Level Architecture Overview

At a high level, the system is composed of four primary layers:

- **Frontend Layer (Client Side):** Provides the user interface for students, faculty, and administrators.
- **Backend Layer (Application Server):** Handles business logic, API processing, and authentication.
- **Database Layer:** Stores persistent data such as users, sessions, attendance, and feedback.
- **External Services Layer:** Includes integrations like email notifications and external platforms.

This layered structure ensures separation of concerns, improving maintainability and scalability.

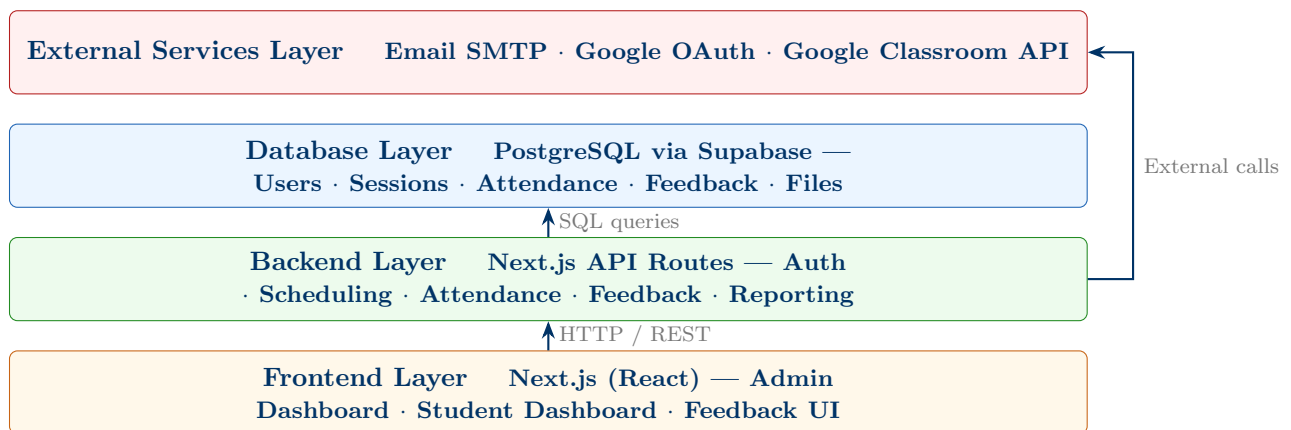


Figure 4.1: Four-layer high-level architecture of CIPD 360 ERP

4.2 Technology Stack

The system is built using a modern web stack:

- **Frontend:** Next.js for efficient rendering and interactivity
- **Backend:** API routes within Next.js for unified development
- **Database:** PostgreSQL (via Supabase) for reliable data storage
- **Authentication:** JWT based stateless authentication
- **Attendance System:** Wi Fi based device detection
- **Integrations:** OAuth support for external platforms

This stack balances performance, scalability, and development efficiency.

4.3 Architectural Components and Responsibilities

The system is divided into modular components:

- **Authentication:** Manages user identity and access control
- **Scheduling:** Handles session planning and updates
- **Attendance:** Processes network data into attendance records
- **Feedback:** Manages feedback collection and analytics
- **Dashboard and Reporting:** Aggregates and visualizes system data
- **Coursework:** Handles materials, assignments, and grading

These components interact through well defined APIs, ensuring modularity.

4.4 Data Flow and Processing Architecture

The system follows a structured data flow:

1. User actions on the frontend trigger API requests
2. Backend processes requests using business logic
3. Data is stored or retrieved from the database
4. Responses are returned and rendered on the frontend

5. Certain events trigger automated workflows

Key automated workflows include:

- **Attendance Processing:** Network data is analysed to generate attendance records
- **Session Lifecycle Handling:** Sessions transition through states (scheduled → ongoing → completed), triggering dependent processes
- **Feedback Activation:** Feedback forms are enabled automatically after session completion

This approach reduces manual effort and ensures consistency across modules.

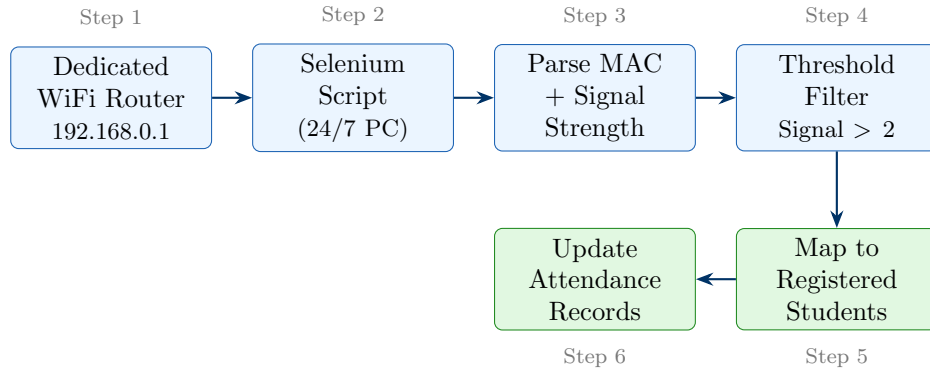


Figure 4.2: WiFi-based automated attendance detection pipeline

4.5 Integration of Attendance Pipeline

A key architectural feature of the system is the integration of the attendance pipeline with the core application.

The process involves:

- Collection of Wi Fi snapshot data from the network
- Mapping detected devices to registered users
- Processing detection data into attendance records
- Storing both raw and processed data for analysis

This pipeline operates alongside the main application and continuously feeds processed attendance data into the database. The stored data is then utilized by dashboards, reporting modules, and analytics systems.

By integrating this pipeline directly into the architecture, the system enables automated, scalable, and data driven attendance tracking.

4.6 Database Design Overview

The system uses a relational database (PostgreSQL) to manage structured data across modules. Core entities include:

- Users and roles
- Sessions and scheduling data
- Attendance records and device mappings
- Feedback forms and responses
- Assignments and submissions

Relationships ensure referential integrity and efficient querying. The schema is normalized to reduce redundancy, with selective denormalization for faster analytics queries.

4.7 API Design and Communication

The system follows an API driven architecture for frontend backend interaction.

Key characteristics:

- **Role Based Endpoints:** Separate APIs for admin, faculty, and students
- **Stateless Communication:** JWT based authentication
- **Structured Responses:** Consistent and enriched data formats
- **Error Handling:** Standardized responses for reliability

This design ensures extensibility and supports future integrations.

4.8 Frontend Implementation

The frontend is developed using Next.js with a focus on usability and performance.

Key aspects include:

- Component based design for consistency
- Dynamic data fetching via APIs
- Responsive interfaces across devices
- State management for smooth interaction

4.9 Integration of System Components

The system ensures seamless interaction between modules:

- Scheduling drives attendance and feedback workflows
- Attendance data feeds dashboards and reporting
- Feedback data contributes to analytics and evaluation

This integration enables cohesive system behaviour across all components.

Chapter 5

Module wise Design

5.1 Authentication and User Management

The Authentication and User Management module forms the foundational layer of the CIPD 360 ERP system. It is responsible for establishing user identity, enforcing access control, and enabling secure interaction across all system components. As all modules depend on user specific behavior, this module acts as the entry point and control boundary for the system.

5.1.1 Role Based Access Control

The system is structured around two primary roles:

- **Administrator (Admin):** Full access to scheduling, attendance, reporting, and configuration. Along with managing faculties and other supporting staff.
- **Student:** Access to schedules, learning materials, feedback submission, and attendance tracking.

Each role is associated with defined permissions, ensuring secure and relevant access while preventing unauthorized operations.

5.1.2 Authentication Mechanism

The system uses a secure login mechanism based on email or enrollment number with password verification.

Key features include:

- Secure password storage using hashing (bcrypt)
- JWT based authentication for validating API requests

- Stateless session handling through token based authorization
- Logout and session expiry for improved security

This approach ensures secure and scalable authentication across the system.

5.1.3 Role Scoped Session Management

To support multiple roles efficiently, the system maintains separate session tokens for different user roles. This allows simultaneous access across roles without conflicts and ensures proper isolation of role specific data.

5.1.4 User Profile and Settings Management

Users can manage their profiles and preferences through dedicated interfaces.

Key functionalities include:

- Viewing personal and academic information
- Managing preferences such as notifications and interface settings
- Secure password updates through verification mechanisms

The system stores preferences in a flexible format, allowing future extensibility.

5.1.5 Device Identity Registration (MAC Binding)

The module integrates device registration to support automated attendance tracking.

- Students register the MAC address of their primary device
- The address is validated and stored, pending administrative verification
- Verified devices are used for WiFi based attendance detection

This establishes a secure link between user and device identity while minimizing misuse.

5.1.6 External Integration Support (OAuth)

The system supports OAuth based integration with external platforms such as Google Classroom.

- Users can link external accounts securely
- Tokens are stored for future API interactions
- External academic data can be integrated into the ERP

5.1.7 Security Considerations

Security is ensured through hashed passwords, token based authentication, role based access control, input validation, and controlled device verification for attendance integrity.

5.2 Scheduling System

The Scheduling System is a core operational module responsible for managing the lifecycle of academic sessions. It defines when and how teaching activities occur and drives downstream processes such as attendance tracking, feedback collection, and reporting.

5.2.1 Session Creation and Management

Administrators can create and manage sessions with attributes such as title, course, faculty, date, time, duration, skills, and venue. Optional metadata such as session type can also be defined.

The system supports both manual and recurring scheduling, allowing multiple sessions per day. Once created, sessions are automatically reflected across dashboards for students and faculty.

5.2.2 Student Centric Schedule View

Students are provided with a personalized scheduling interface:

- Calendar based views (day/week/month)
- Course based filtering for relevance
- Detailed session information including instructor, timing, and venue

This ensures clear visibility and easy navigation of schedules.

5.2.3 Admin Scheduling Console

Administrators manage schedules through a dedicated interface with:

- Session listing and filtering
- List and calendar views
- Create/edit workflows with validation
- Metadata management for session classification

The interface is designed for efficient and structured scheduling operations.

5.2.4 Integration with Other Modules

The Scheduling System coordinates workflows across the ERP:

- Drives attendance tracking based on session timing
- Triggers feedback collection after session completion
- Contributes to dashboards and reporting analytics

This integration ensures that scheduling acts as a central driver of system automation.

5.3 Attendance System

The Attendance System is one of the most critical and technically distinctive components of the CIPD 360 ERP. Unlike traditional attendance mechanisms, this system is designed as an automated, data driven pipeline that leverages WiFi network signals to infer student presence. It minimizes manual intervention, reduces the possibility of proxy attendance, and provides both real time and analytical insights into student participation.

5.3.1 Overview of Attendance Approach

The system adopts a WiFi based attendance detection mechanism, where student presence is determined by detecting their registered device on a controlled network.

The core idea is based on the following principles:

- Each student registers a unique device (via MAC address)
- The system monitors devices connected to the CiPD network
- Periodic detections are used to infer presence during a session
- Attendance is computed based on detection frequency

This approach ensures that attendance tracking is passive, scalable, and tightly integrated with the existing infrastructure.

5.3.2 Device Registration and Identity Mapping

Attendance automation relies on establishing a reliable mapping between users and their devices.

- Each student registers the MAC address of their primary device through the system
- The MAC address is validated and stored in the database

- Administrative approval is required to verify the device
- Only verified devices are considered for attendance tracking

This verification step is essential to prevent misuse and ensure that attendance records are linked to legitimate user identities.

5.3.3 WiFi Detection Mechanism

The system continuously monitors the CiPD network to identify connected devices and infer student presence. This is achieved through an automated detection pipeline that operates independently of user interaction.

The detection process is implemented using an autonomous script running continuously (24/7) on a dedicated PC connected to the CiPD network within the training room. The script utilizes browser automation (Selenium) to perform periodic data extraction from the network router interface (accessible via the local domain 192.168.0.1). This allows the system to retrieve real-time information about all devices currently connected to the network.

The detection workflow involves:

- **Capturing periodic WiFi snapshots:** The script collects data at regular intervals, listing all connected devices
- **Extracting device identifiers:** MAC addresses and associated signal strength values are parsed from the router interface
- **Device mapping:** Detected MAC addresses are matched with registered student devices stored in the system
- **Signal strength filtering:** Devices are evaluated based on signal strength to ensure proximity based detection

To improve reliability and prevent misuse, multiple experiments were conducted to determine an optimal signal strength threshold. Based on these observations, only devices with a signal strength greater than a defined threshold (e.g., >2) are considered valid for attendance. Additionally, we have modified channel width and frequency to ensure attendance is marked only for those students who are actually present in the CiPD training room. This ensures that students who are connected to the network from distant locations are not incorrectly marked as present.

After validating signal strength and mapping device identities, the processed data is used to update attendance records for each detection interval. These periodic detections collectively form the raw dataset used for computing final attendance.

This approach enables a passive, automated, and scalable mechanism for attendance tracking while maintaining reasonable safeguards against proxy or remote attendance.

5.3.4 Session Based Attendance Window

Attendance is evaluated within the time boundaries of each scheduled session.

For a given session:

- A detection window is defined from session start time to session end time
- Periodic scans (running of script) are conducted at fixed intervals (Every 3 mins)
- Each scan records which registered devices are detected and updates this in database

This results in multiple detection points across a session, forming a temporal presence pattern for each student.

5.3.5 Attendance Decision Logic

Attendance is determined based on the frequency and consistency of device detections during a session. The system operates on periodic network scans conducted every 3 minutes, resulting in multiple detection points across the duration of a class.

For each session, the total number of detections (pings) for a student is compared against the expected number of scans to compute a presence percentage. This ensures that attendance is based on sustained presence rather than a single detection.

The decision criteria include:

- Evaluation of detections distributed across the entire session duration
- Consideration of both presence consistency and timing of entry
- Classification of attendance based on percentage of successful detections

Additionally, an early presence factor is considered, where students detected within the initial minutes of the session are rewarded, encouraging timely attendance.

This approach ensures that attendance reflects actual participation over time and reduces the possibility of proxy or short-duration presence being misclassified as full attendance.

5.3.6 Points Based Attendance Scoring

To provide a more granular representation of participation, the system uses a points-based scoring model instead of a simple present/absent classification.

Attendance is scored out of a total of 5 base points, with an additional bonus point awarded for timely arrival.

Scoring Logic:

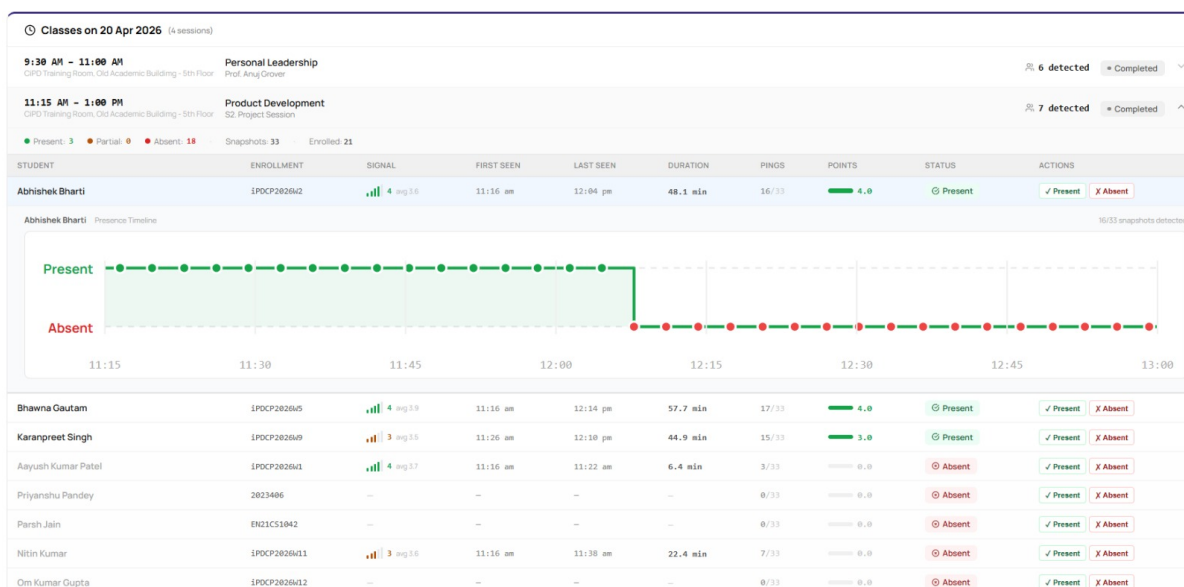


Figure 5.1: Session-level attendance detail showing per-student presence timeline, pings, signal strength, and computed points

- **Base Score (5 points):** Assigned based on overall presence during the session
- **Bonus Point (+1):** Awarded if the student's first detection occurs within 8 minutes of session start time

Attendance Calculation (based on % of detections):

- $\geq 85\%$ detections: Full attendance $\rightarrow$ 5 points
- $70\% \leq$ detections $< 85\%$: Slightly reduced $\rightarrow$ 4 points
- $50\% \leq$ detections $< 70\%$: Partial attendance $\rightarrow$ 3 points
- $< 50\%$ detections: Marked absent $\rightarrow$ 0 points

This model ensures that attendance scoring accounts for both duration of presence and punctuality, providing a more accurate measure of student engagement. It also enables richer analytics compared to a binary attendance system, supporting better evaluation of participation patterns.

5.3.7 Real Time Monitoring and Admin Controls

The system provides a comprehensive interface for administrators to monitor attendance in real time.

Key features include:

- **Live Student Detection:** Displays currently detected devices during ongoing sessions

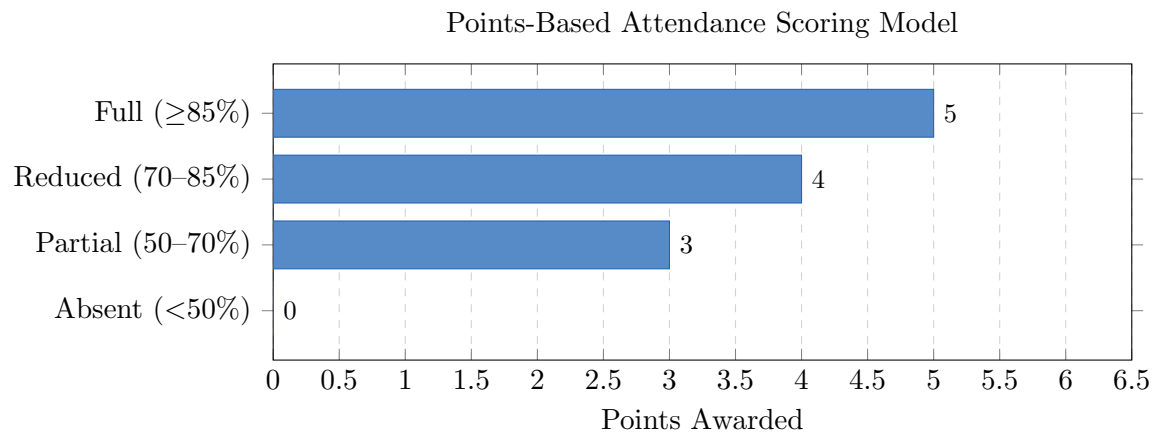


Figure 5.2: Points-based attendance scoring thresholds and corresponding values

- **Session Level Analytics:** Shows number of detected students, total enrolled students, and session status. Can also view individual entry and exit time of students in each session.
- **Per Student Breakdown:** Detailed view including detection timestamps, signal strength, duration, and computed points
- **Manual Override:** Administrators can mark a student as present or absent in exceptional cases
- **Penalty Enforcement:** In cases of suspected misuse, penalties can be applied to discourage proxy attendance

These controls ensure that the system remains flexible and allows human intervention when required.

AD
MAIN
ANALYTICS
SYSTEM

Connected Students (8 students)
Search name, enrollment, MAC

STUDENT	ENROLLMENT	SIGNAL	DEVICE	IP ADDRESS	DURATION	STATUS
Test Student	TEST001	0/5 Weak	DESKTOP-T5FJ3IE	192.168.0.230	1 Day 4 Hours 23 Minutes	Connected
Karanpreet Singh	1PDCP2026W9	4/5 Strong	Redmi-Note-11	192.168.0.225	18 Minutes	Connected
Parsh Jain	EN21CS1042	4/5 Strong	Parshs-iPhone	192.168.0.131	0 Minutes	Connected
Sayem Azam	1PDCP2026W16	3/5 Good	PROFESSOR	192.168.0.170	2 Hours 41 Minutes	Connected
Bhawna Gautam	1PDCP2026W5	4/5 Strong	iPhone13	192.168.0.133	22 Minutes	Connected
Divyanshu Mohan Mani Tripathi	1PDCP2026W8	4/5 Strong	OnePlus-Nord2-5G	192.168.0.239	5 Minutes	Connected
Balvinder Kumar	1PDCP2026W4	4/5 Strong	POCO-M4-Pro-5G	192.168.0.134	0 Minutes	Connected
Shreshth Yadav	1PDCP2026W17	3/5 Good	POCO-X5-5G	192.168.0.247	1 Minute	Connected

Unidentified Devices (2 devices — MAC not registered to any student)

MAC ADDRESS	DEVICE NAME	SIGNAL	IP ADDRESS	DURATION
04:CF:4B:A9:6C:A3	PJ	5/5	192.168.0.198	26 Minutes
2A:59:CC:DA:8D:95	POCO-M5	4/5	192.168.0.173	19 Minutes

Figure 5.3: Real-time live detection view showing connected students with signal strength, device name, IP address, duration, and unidentified devices

5.3.8 Data Storage and Processing

The attendance system maintains multiple layers of data for accuracy and auditability:

- **Raw Data:** WiFi snapshots and ping logs containing detected devices
- **Processed Data:** Aggregated attendance records per student per session
- **Metadata:** Detection timestamps, signal strength, and duration of presence

Attendance records are updated dynamically and stored in a structured format, enabling both real time access and historical analysis.

5.3.9 Automation and Background Processing

To ensure reliability, attendance processing is supported by automated background workflows.

- Periodic processes analyze WiFi snapshots and compute attendance
- Sessions are automatically marked as completed after their scheduled duration
- Attendance records are finalized without requiring manual intervention

This reduces dependency on manual actions and ensures consistency in attendance data.

5.3.10 Challenges and Limitations

While the system provides significant advantages, it also introduces certain challenges:

- Dependence on network availability and scanner reliability
- Potential inaccuracies due to weak signal strength or device disconnection
- Requirement for proper MAC registration and verification

5.4 Feedback System

The Feedback System is designed to capture structured student feedback for each lecture and transform it into actionable insights for faculty and administrators. It serves as the qualitative intelligence layer of the CIPD 360 ERP, using data to generate insights and actionable actions.

5.4.1 Session Based Feedback Model

The system follows a session driven feedback approach, where feedback forms are generated for each completed session.

Key characteristics:

- Feedback is linked directly to individual sessions
- Only eligible students (typically those associated with the session) can submit responses
- Feedback availability is time bound, with defined deadlines

This ensures that feedback is contextual, relevant, and collected shortly after the session, improving response quality and accuracy.

5.4.2 Feedback Collection Workflow

The feedback process is tightly integrated with the session lifecycle:

1. A session is marked as completed by the system or administrator
2. The system triggers feedback availability for relevant students
3. Students receive notifications regarding pending feedback
4. Students access the feedback interface and submit responses
5. Responses are stored and made available for analysis

This workflow ensures minimal manual intervention and maintains consistency across all sessions.

5.4.3 Feedback Form Structure

The system supports multiple types of feedback questions to capture both quantitative and qualitative responses.

Supported question formats include:

- **Rating Scale (e.g., 1–5):** Used for evaluating aspects such as content quality, structure, and delivery
- **Yes/No Questions:** Used for binary assessments, such as whether a session started on time
- **Conditional Questions:** Follow up questions triggered based on previous responses
- **Descriptive/Text Responses:** Allow students to provide detailed feedback and suggestions

This flexible structure enables comprehensive evaluation of sessions while maintaining a simple user experience.



Figure 5.4: Student feedback submission interface for a completed session

5.4.4 Feedback Submission Constraints

To encourage higher participation rates, the system is designed for quick and efficient feedback submission.

- Feedback forms are optimized to be completed within a short duration (approximately one minute)
- Default selections or simplified input mechanisms are used to reduce friction
- Time bound availability ensures that feedback is submitted promptly

These design choices aim to balance usability with the need for meaningful data collection.

5.4.5 Notification and Reminder System

The system incorporates a notification driven approach to ensure timely feedback submission.

Key features include:

- **Initial Notification:** Students are notified when feedback becomes available for a session
- **Reminder Notifications:** Students who have not submitted feedback receive reminders before the deadline
- **Administrative Control:** Administrators can trigger additional reminders if required

This approach improves submission rates and ensures that feedback data is sufficiently comprehensive.

5.4.6 Feedback Analytics and Insights

Collected feedback is processed and presented through analytical dashboards for administrators and faculty.

Key analytics include:

- **Average Ratings:** Aggregated scores for different evaluation parameters
- **Response Distribution:** Breakdown of ratings across students
- **Submission Rates:** Percentage of students who submitted feedback for a session
- **Session Level Insights:** Comparative analysis across sessions, courses, or instructors
- **Descriptive Feedback Review:** Access to textual responses for qualitative insights

These analytics help identify patterns, strengths, and areas for improvement in teaching and session management.

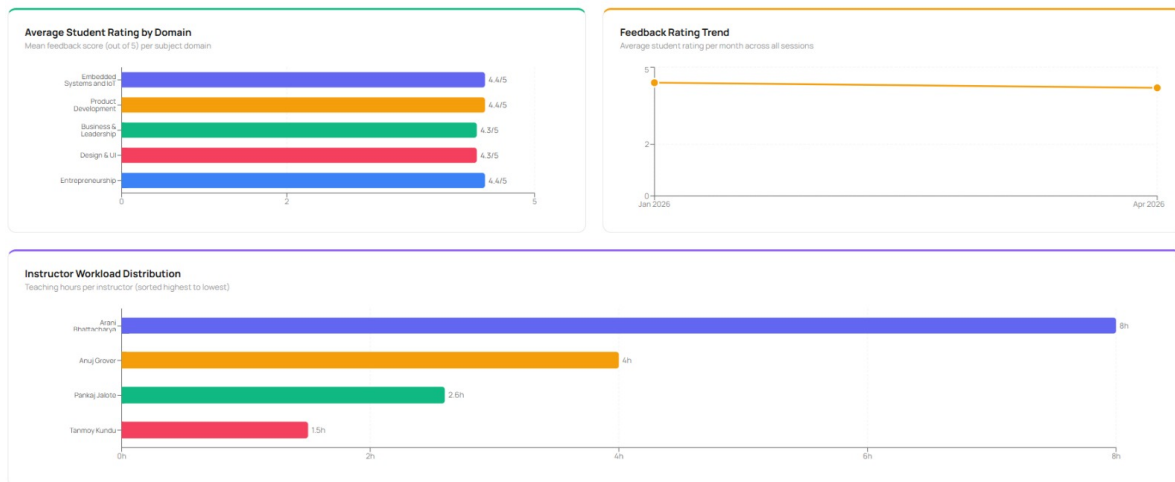


Figure 5.5: Feedback analytics dashboard showing average ratings by domain, feedback rating trend, and instructor workload distribution

5.4.7 Design Considerations and Trade offs

Several design decisions were made to balance usability and data quality:

- **Notification Based Eligibility:** Feedback visibility is controlled through notifications, ensuring targeted distribution
- **Quick Submission Design:** Simplified forms increase participation but may reduce depth of responses

- **Time Bound Feedback Window:** Improves relevance but may lead to missed submissions if deadlines are too strict
- **Privacy Considerations:** Access to individual student responses is controlled to maintain confidentiality

These trade offs were carefully considered to ensure a practical and effective feedback system.

5.5 Dashboard and Reporting

The Dashboard and Reporting module provides a unified interface for visualizing system data and deriving actionable insights. It serves as the presentation and analytics layer of the CIPD 360 ERP, consolidating information from scheduling, attendance, and feedback into role specific views. This enables users to monitor performance, track engagement, and make informed decisions.

5.5.1 Role Based Dashboard Design

The system implements dashboards tailored to different user roles:

- **Administrator Dashboard:** Offers an overview of system activity, including attendance trends, session statistics, feedback summaries, and faculty workload, supporting monitoring and decision making.
- **Student Dashboard:** Displays personalized information such as upcoming sessions, attendance percentage, pending feedback, learning resources, and assignments.

Dashboards are dynamically populated based on user roles, ensuring relevance and clarity.

5.5.2 Key Dashboard Components

Dashboards consist of components that provide both summary and detailed insights:

- **Cards:** Display key metrics such as attendance percentage, session counts, and pending tasks
- **Graphs and Visualizations:** Show trends in attendance, feedback, and participation
- **Activity Panels and Alerts:** Highlight recent updates and notify users of pending actions

These elements provide a balance between overview and actionable information.

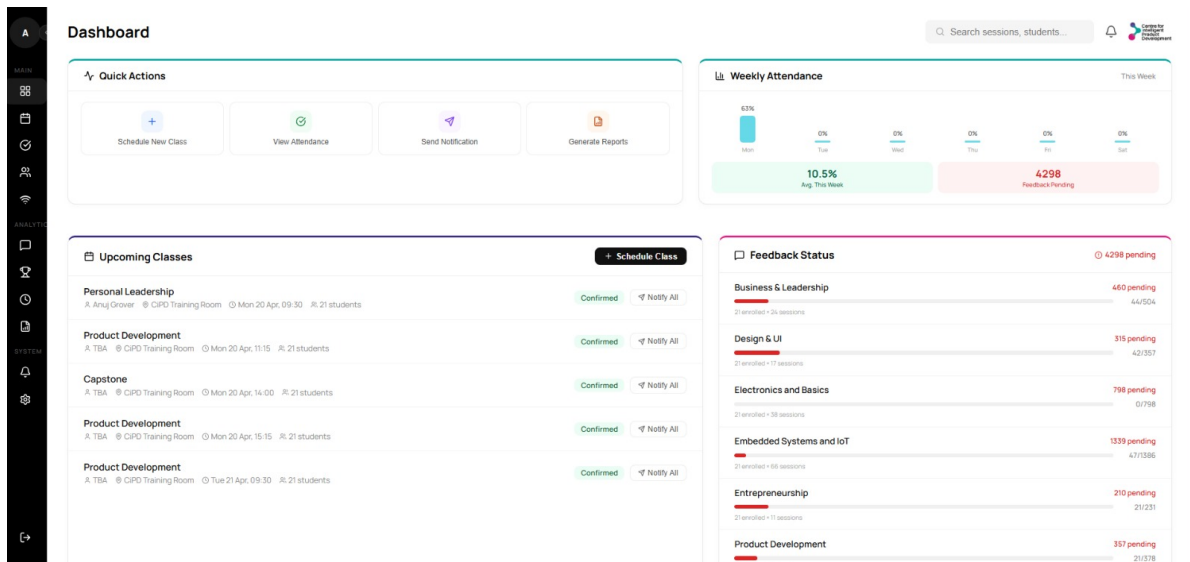


Figure 5.6: Admin dashboard showing quick actions, weekly attendance chart, upcoming classes list, and per-course feedback status

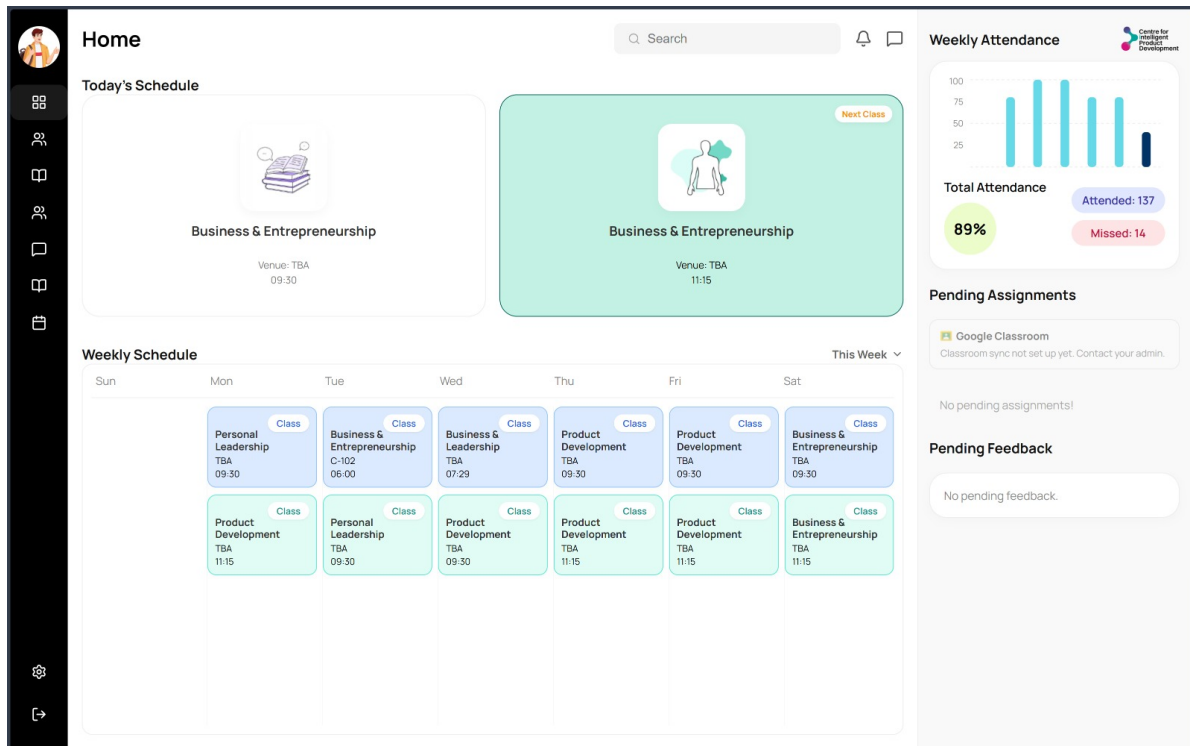


Figure 5.7: Student home dashboard showing today's schedule, weekly calendar view, total attendance (89%), pending assignments, and pending feedback

5.5.3 Analytics and Insights

The reporting system generates insights across multiple dimensions:

- **Attendance Analytics:** Includes time based trends, course wise analysis, and student level participation metrics

- **Feedback Analytics:** Covers average ratings, response distributions, submission rates, and qualitative responses
- **Faculty and Session Analytics:** Tracks sessions conducted, teaching hours, and engagement levels

These analytics help identify patterns, evaluate performance, and support decision making.

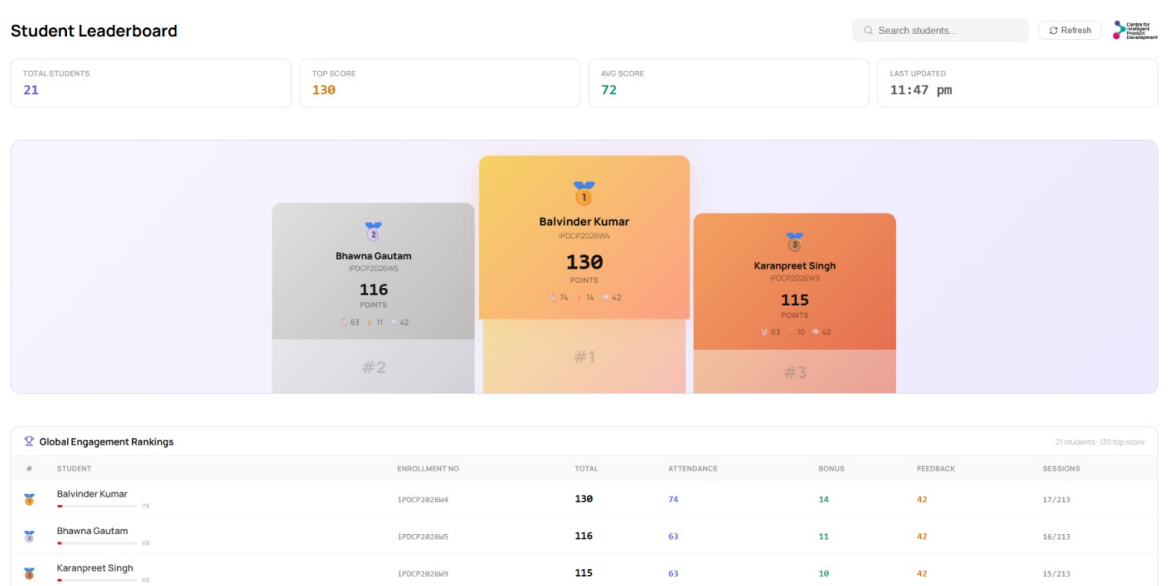


Figure 5.8: Student engagement leaderboard showing global rankings by total points, attendance score, bonus points, feedback score, and sessions attended

5.5.4 Reporting and Data Export

The system supports generation of structured reports for administrative use:

- Report types include attendance, feedback, and session analytics
- Export options such as CSV and PDF are supported
- Filters allow customization based on date, course, or user group

This functionality enables efficient data sharing and external analysis.

5.6 Content and Assignment System

5.6.1 Course Centric Content Management

The system organizes academic content in a course-centric manner, where materials are associated with specific sessions and courses. A total of 10 modules/courses have been implemented:

- Business & Entrepreneurship
- Entrepreneurship
- Business & Leadership
- Personal Leadership
- Electronics and Basics
- Capstone
- Embedded Systems and IoT
- Product Development
- Software & App Development
- Design & UI

Key features include:

- **Lecture Material Upload:** Faculty and administrators can upload resources such as slides, PDFs, reference links, and notes
- **Session-Based Association:** Materials are linked to specific sessions for contextual access
- **Centralized Access:** Students can access all materials through a unified interface

Additionally, the system incorporates a structured skills framework with 100+ engineering and product development skills across 46 categories, enabling better alignment of coursework with learning outcomes.

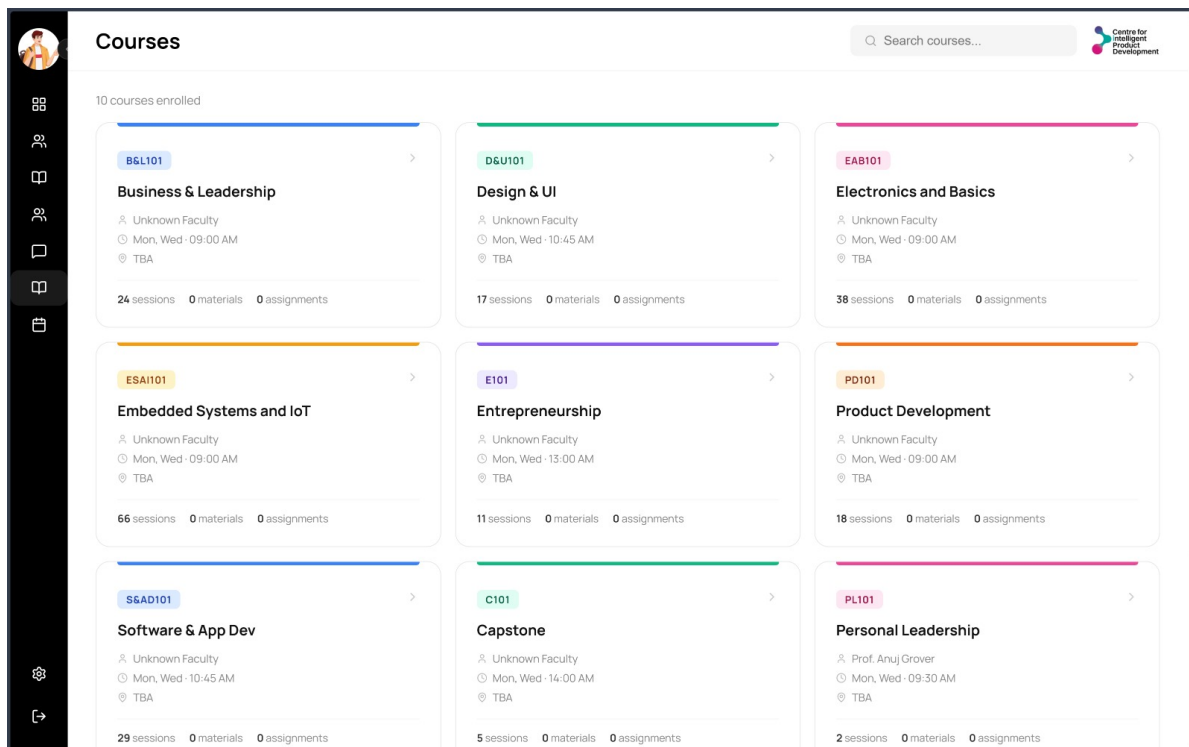


Figure 5.9: Student courses page showing all 10 enrolled modules with session counts, materials, and assignments

5.6.2 Role Based Content Access

Access to content is governed by user roles:

- **Faculty/Admin:** Can upload, update, and manage content and assignments
- **Students:** Can view and download materials

This ensures content integrity while maintaining ease of use.

5.6.3 Assignment Management and Submissions

The system provides a structured workflow for assignment management:

- **Assignment Creation:** Faculty can define assignments with title, description, and deadlines
- **Course/Session Mapping:** Assignments are linked to relevant courses or sessions
- **Student Access:** Assignments are accessible via dashboards and course pages

For submissions:

- **File Upload Support:** Students can submit documents such as PDFs or reports
- **Submission Metadata:** The system records timestamps, student identity, and version details
- **Version Handling:** Multiple submissions are supported before deadlines

Faculty can evaluate submissions by assigning marks and providing feedback, which is visible to students.

5.6.4 Integration with Google Classroom

The system integrates with Google Classroom using OAuth-based authentication.

- Assignments created in Google Classroom are automatically fetched and displayed within the ERP
- Students can view both ERP and Google Classroom assignments in a single dashboard
- Eliminates the need to switch platforms while ensuring consistency

This integration extends functionality without duplicating existing systems and improves overall usability.

5.6.5 Dashboard Integration

The module is integrated with the dashboard to enhance visibility:

- Students can track pending assignments and upcoming deadlines
- Faculty can monitor submission status and progress

This ensures users remain informed and improves task management efficiency.

5.6.6 Role in Overall System

The Content and Assignment System supports:

- Delivery of learning resources
- Management of coursework
- Tracking of academic performance

By integrating with scheduling, attendance, and feedback modules, it contributes to a cohesive and comprehensive academic management system.

Chapter 6

Results and Observations

6.1 System Outputs

The system successfully provides a unified interface for managing academic operations. Key outputs include:

- **Role Based Dashboards:** Personalized views displaying schedules, attendance, and pending tasks.
- **Attendance Records:** Automated session wise and student wise tracking, including participation scores.
- **Feedback Data:** Structured responses aggregated into summaries and analytics.
- **Reports and Visualizations:** Data presented through graphs, tables, and downloadable reports.

6.2 Observations

Based on system behaviour and usage, several key observations were made:

- **Reduction in Manual Effort:** Automation of attendance tracking and feedback collection significantly reduces administrative workload.
- **Improved Data Availability:** Real time access to scheduling, attendance, and feedback data improves transparency for all stakeholders.
- **Enhanced Student Engagement Tracking:** The combination of attendance and feedback data provides a more comprehensive view of student participation.
- **Streamlined Workflows:** Integration between modules ensures that processes such as attendance computation and feedback rollout occur without manual intervention.

- **Centralized Data Management:** Consolidation of academic data into a single system simplifies monitoring and reporting.

These observations indicate that the system effectively addresses many of the inefficiencies present in traditional approaches.

6.3 Limitations

Despite its effectiveness, the system has certain limitations:

- **Dependence on Network Infrastructure:** The attendance system relies on WiFi connectivity and accurate device detection, which may be affected by network conditions.
- **Device Dependency:** Attendance accuracy depends on students connecting their registered devices to the network.
- **Potential Security Concerns:** Although mitigated through verification, MAC based systems can be susceptible to spoofing.
- **Limited Scale Testing:** The system has primarily been tested within the CiPD environment and may require further validation for large scale deployment.

These limitations highlight areas where further refinement and enhancement may be required.

Chapter 7

Challenges and Solutions

7.1 Attendance Reliability and Accuracy

Challenge:

Ensuring reliable attendance detection using WiFi signals was a major challenge. Factors such as device disconnections, weak signal strength, and intermittent connectivity could lead to inaccurate presence detection.

Solution:

Instead of relying on a single detection point, a multi interval detection strategy was implemented. Attendance is computed based on repeated detections across the session duration, reducing the impact of temporary disconnections. Additionally, signal strength filtering and threshold based logic were used to improve detection accuracy.

7.2 Network Access and Infrastructure Constraints

Challenge:

Initial attempts to use the institute's existing WiFi network for attendance tracking were restricted due to permission limitations. Continuous monitoring of connected devices was not allowed.

Solution:

A dedicated WiFi network was established specifically for the CiPD program. A separate router was configured to provide controlled access, enabling reliable detection of connected devices without violating network policies. This approach provided full control over the attendance data pipeline.

7.3 Hardware and Firmware Limitations

Challenge:

To enhance network level data collection, attempts were made to install custom firmware (Open-WRT) on the router. However, the specific hardware used did not support this modification.

Solution:

The system was adapted to work within the constraints of the existing router firmware. Instead of deep packet level analysis, the implementation relied on available device detection capabilities, ensuring functionality without requiring unsupported configurations.

7.4 Preventing Proxy Attendance

Challenge:

A key concern in automated attendance systems is the possibility of proxy attendance, especially when relying on device based identification.

Solution:

A combination of safeguards was implemented, including:

- MAC address registration with administrative verification
- Monitoring of detection patterns for anomalies
- Provision for manual checks and penalties

These measures reduce the likelihood of misuse while maintaining system usability.

7.5 Feedback Participation and Engagement

Challenge:

Ensuring consistent student participation in feedback submission is often difficult, leading to incomplete or biased data.

Solution:

The feedback system was designed to be lightweight and quick to complete. A notification and reminder mechanism was implemented to prompt students to submit feedback within a defined time window, improving participation rates.

7.6 Balancing Automation and Control

Challenge:

While automation reduces manual effort, it also introduces risks if processes fail or produce incorrect results without oversight.

Solution:

Administrative controls such as manual overrides, monitoring dashboards, and configurable parameters were incorporated. This ensures that the system remains flexible and allows human intervention when necessary.

Chapter 8

Future Scope

While the CIPD 360 ERP system provides a functional and integrated solution for academic management, there are several opportunities for further enhancement and expansion.

One potential improvement is the incorporation of advanced analytics and machine learning techniques to derive deeper insights from attendance and feedback data. Predictive models could be used to identify at risk students, analyze engagement trends, and support data driven academic interventions.

The attendance system can be further strengthened by exploring hybrid approaches, such as combining WiFi based detection with technologies like Bluetooth or facial recognition. This would improve accuracy and reduce dependency on a single method. But a more robust approach would be to integrate this software over to mobile S and utilize the institution network for a full proof mechanism.

Another area of expansion is the development of a dedicated mobile application, which would enhance accessibility and enable real time notifications, making the system more convenient for daily use.

The platform can also be extended to support full scale Learning Management System (LMS) capabilities, including advanced assignment workflows, peer evaluations, and incorporation of agentic AI to automate workflows.

From a deployment perspective, the system can be scaled beyond the CiPD program to support multiple courses, departments, or institutions, with configurable modules to adapt to different academic structures. Meaning it can be cross institutionalized and can be used to support similar programs in other colleges.

Chapter 9

Conclusion

This project presented the design and development of CIPD 360 ERP, an integrated academic management system tailored for the CiPD program at IIIT Delhi. The system was developed to address key inefficiencies in traditional academic workflows by introducing automation, centralization, and data driven insights.

By integrating scheduling, attendance tracking, feedback collection, and reporting into a unified platform, the system successfully streamlines core academic processes. A significant contribution of this work is the implementation of a WiFi based automated attendance mechanism, which demonstrates a practical and scalable alternative to conventional attendance systems. In addition, the incorporation of a structured feedback system and role based dashboards enables continuous monitoring and evaluation of academic activities.

The project also highlights the importance of iterative development and user centric design. Through continuous feedback from stakeholders and multiple rounds of experimentation, the system was refined to align with real world requirements and constraints. Challenges related to network limitations, reliability, and system integration were addressed through thoughtful design decisions and practical trade offs.

Overall, the CIPD 360 ERP system demonstrates the feasibility of building a scalable and modular academic ERP solution for dynamic educational environments. While certain limitations remain, the system establishes a strong foundation for future enhancements and broader deployment.

Appendix A

API Endpoints

A.1 Authentication APIs

Method	Endpoint	Description
POST	/api/auth/login	User login (email/enrollment + password)
POST	/api/auth/signup	Student registration
GET	/api/auth/me	Get current user profile
POST	/api/auth/logout	Logout user
POST	/api/auth/update-password	Change password
GET	/api/auth/google/connect	Initiate Google OAuth
GET	/api/auth/google/callback	Handle OAuth callback

Table A.1: Authentication API endpoints

A.2 User Management APIs

Method	Endpoint	Description
GET	/api/students/profile	Fetch student profile
GET	/api/students/settings	Get user preferences
PATCH	/api/students/settings	Update preferences
PATCH	/api/students/mac	Register MAC address

Table A.2: User management API endpoints

Method	Endpoint	Description
<i>Student</i>		
GET	/api/calendar/sessions	Fetch sessions by date range
GET	/api/students/schedule/today	Today's schedule
GET	/api/students/schedule/week	Weekly schedule
<i>Admin</i>		
GET	/api/admin/schedule	Get all sessions (filtered)
GET	/api/admin/sessions	List sessions
POST	/api/admin/sessions	Create session
PATCH	/api/admin/sessions/[id]	Update session / mark completed
GET	/api/admin/session-types	Fetch session types
POST	/api/admin/session-types	Create session type
GET	/api/admin/lookup	Fetch courses, faculty, venues, etc.
GET	/api/admin/skills	Get skills
POST	/api/admin/skills	Create skill
GET	/api/admin/categories	Get categories
POST	/api/admin/categories	Create category
POST	/api/admin/sessions/rollout-feedback	Trigger feedback rollout

Table A.3: Scheduling and session API endpoints

Method	Endpoint	Description
<i>Student</i>		
GET	/api/students/attendance/summary	Attendance overview
GET	/api/students/attendance/sessions	Session-wise attendance
GET	/api/students/attendance/presence	Live presence status
POST	/api/students/attendance/ping	Send attendance ping
<i>Admin</i>		
GET	/api/admin/attendance/weekly	Weekly attendance report
GET	/api/admin/attendance/sessions-by-date	Sessions for a date
GET	/api/admin/attendance/session-students	Detailed student attendance
GET	/api/admin/attendance/snapshot	Snapshot-based attendance
POST	/api/admin/attendance/override	Manual attendance override
<i>Monitoring</i>		
GET	/api/admin/live-students	Real-time connected devices
GET	/api/admin/wifi-logs	Wi-Fi logs for audit
<i>Settings</i>		
GET	/api/admin/settings/config	Get system config
PUT	/api/admin/settings/config	Update config
GET	/api/admin/settings/mac-approvals	Pending MAC approvals
PATCH	/api/admin/settings/mac-approvals	Approve/reject MAC
GET	/api/admin/settings/bssid	Get router BSSID
POST	/api/admin/settings/bssid	Add BSSID
PATCH	/api/admin/settings/bssid	Update BSSID
DELETE	/api/admin/settings/bssid	Delete BSSID

Table A.4: Attendance API endpoints (Student, Admin, Monitoring, and Settings)

A.3 Scheduling & Session APIs

A.4 Attendance APIs

A.5 Feedback APIs

Method	Endpoint	Description
<i>Student</i>		
GET	/api/feedback/pending	Get pending feedback
POST	/api/feedback/submit	Submit feedback
GET	/api/feedback/my-response	View submitted response
GET	/api/feedback/leaderboard	Engagement leaderboard
<i>Admin</i>		
GET	/api/admin/feedback/analytics	Feedback analytics
GET	/api/admin/feedback/forms	List feedback forms
PATCH	/api/admin/feedback/forms	Update deadlines
GET	/api/admin/feedback/questions	Get questions
POST	/api/admin/feedback/questions	Add question
PATCH	/api/admin/feedback/questions	Update question
DELETE	/api/admin/feedback/questions	Delete question
GET	/api/admin/feedback/student-response	View individual response
GET	/api/admin/feedback/status	Pending feedback status

Table A.5: Feedback API endpoints (Student and Admin)

A.6 Notification APIs

Method	Endpoint	Description
POST	/api/admin/notifications	Send notifications/reminders
GET	/api/student/notifications	Fetch student notifications
PATCH	/api/student/notifications	Update notification status

Table A.6: Notification API endpoints

A.7 Cron / Automation APIs

Method	Endpoint	Description
GET	/api/cron/process-attendance	Auto process attendance
GET	/api/cron/feedback-reminder	Send feedback reminders

Table A.7: Cron and automation API endpoints

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